

Solinteg EMS BattCtrl

Observed behavior on MHT firmware 9.x and confirmed qEMS control profiles

Updated 2026-08-08 from direct inverter testing, qEMS controller development, standalone qEMS runtime hardening, and independent validation credited in the acknowledgements. Some behavior differs from the official Modbus register descriptions.

Most important correction

Register 50207 controls requested battery charge/discharge power, not total inverter AC power. Positive requests battery discharge; negative requests battery charge. PV is then added or used for charging according to priority and the AC import/export limits.

Overview

EMS BattCtrl lets an external Energy Management System control battery power while the inverter continues to manage battery safety, SOC limits, charge/discharge current limits, PV MPPT, grid synchronization, and conversion limits.

The EMS supplies a battery target plus inverter-side AC input/output limits. The inverter combines PV, battery, house load, and grid power while the battery BMS remains the final authority.

- A controller does not need model-specific BMS permission sensors to remain safe.
- When the BMS limits or refuses power, the controller may continue requesting the target; actual battery power will simply differ.
- qEMS modes described later are external control profiles, not native Solinteg working modes.

Confirmed sign conventions and power balance

Quantity	Positive value	Negative value
BattCtrl target (50207)	Battery discharge	Battery charge
Battery power sensor	Battery discharging	Battery charging
Native inverter grid meter	Site exporting	Site importing
PV power	PV generation	Normally not used

Using the native inverter meter sign above, a useful approximate site power balance is:

House load $\approx$ PV power + battery power - grid power

Demand after PV (D) $\approx$ battery power - grid power $\approx$ house load - PV power

The second equation is especially useful for a feedback controller because it derives the current house-versus-PV deficit without requiring a separate house-load sensor.

Registers

Register	Type	Scale	Observed function
50207	S16	0.01 kW/count	Requested battery charge/discharge power
50208	S16	0.01 kW/count	Maximum positive inverter AC output
50209	S16	0.01 kW/count	Maximum negative inverter AC input
50210	U16 enum	0 = PV, 1 = Battery	Priority when AC capacity or limits constrain PV and battery

Power-register conversion:

Engineering value (kW) = register value x 0.01

Register value = engineering value (kW) x 100

Requested value	Raw register value
10.00 kW	1000
6.50 kW	650
2.00 kW	200
1.00 kW	100
0.00 kW	0
-1.00 kW	-100
-2.00 kW	-200
-10.00 kW	-1000

Observed register behavior

50207 - Battery charge/discharge target

Despite earlier interpretations and the ambiguous integration naming, confirmed testing shows that 50207 requests battery power.

- Positive target: discharge the battery.
- Negative target: charge the battery.
- Zero target: request neither charge nor discharge.

With PV priority, PV remains available independently of the requested battery discharge. During charging, available PV is used first and grid power supplies the remaining requested charge only when AC input is allowed.

The requested value is not a promise of actual battery power. SOC, BMS current, temperature, voltage, inverter power, and the configured AC limits may reduce or block the result.

50208 - Maximum inverter AC output

50208 caps the inverter's positive AC output. It is not a direct utility-meter export limit because the house load is downstream of the inverter output calculation.

Example with PV priority:

PV available = 1 kW
 Battery target = +6 kW
 Max inverter output = 2 kW

- PV contributes approximately 1 kW.
- Battery discharge is reduced to approximately 1 kW.
- Total inverter output is limited to approximately 2 kW.

If available PV alone exceeds the output limit, PV may also be curtailed. With Battery priority, battery power is preserved first and PV is more likely to be curtailed.

50209 - Maximum inverter AC input

50209 caps negative inverter AC input used for battery charging. It does not include the house load, so the utility meter can show more import than the configured limit.

Example:

Battery target = -7 kW
 PV = 3 kW
 House load = 0.7 kW
 Max inverter input = -2 kW

- PV contributes approximately 3 kW to charging.
- The inverter imports up to approximately 2 kW for charging.
- Actual battery charging is approximately 5 kW.
- The house imports an additional approximately 0.7 kW.
- The utility meter therefore shows approximately 2.7 kW import.

50210 - Priority

Priority matters when PV plus requested battery power exceeds inverter AC capacity or an EMS limit.

Priority	Observed behavior
PV (0)	Preserves PV first. Battery contribution is reduced first when output is constrained. During charging, PV is used before grid input.
Battery (1)	Preserves battery contribution first. PV may be curtailed even when PV is available.

qEMS recommendation

Use PV priority for all qEMS profiles. It gives predictable PV-first behavior and avoids unnecessary PV curtailment.

House load and the utility meter

The EMS AC limits apply to the inverter's own AC exchange, not directly to the site's net utility-meter flow.

- House consumption can increase site import beyond the configured maximum inverter input.
- House consumption can reduce site export below the configured inverter output.
- A site-level import/export target must therefore compensate for house load or use live grid feedback.

This distinction does not change the corrected meaning of 50207: the target is battery power, while 50208 and 50209 constrain inverter-side AC exchange.

Terminology warning: official/integration labels such as “Max Grid Export” and “Max Grid Import” are misleading for registers 50208/50209. Observed behavior is Maximum Inverter AC Output / Maximum Inverter AC Input. These registers are not whole-site or utility-meter limits and must not be treated as service-fuse protection.

Safe activation and relatch sequence

Write-order rule

Whenever entering EMS BattCtrl or performing a full relatch, Working Mode Selection must be the first inverter/Modbus write.

1. Pause the old feedback loop so it cannot write a stale target during the transition.
2. Write Working Mode = EMS BattCtrl.
3. Wait until the inverter reports EMS BattCtrl active. Abort if confirmation is not received.
4. Do not rely on a fixed delay. Continue only after the inverter/HA integration confirms EMS BattCtrl active; then write the complete BattCtrl parameter set.
5. Write priority (normally PV).
6. Write maximum inverter AC output (50208).
7. Write maximum inverter AC input (50209).
8. Write the initial battery target (50207).
9. Start or resume the feedback controller.

Writing BattCtrl parameters before the inverter has entered EMS BattCtrl can be ignored by the active control logic. Critically, an early value can still read back exactly as written while not being enforced. Register readback is therefore not proof that a pre-mode write took effect. The safe sequence is always mode first, confirm BattCtrl active, then write priority, inverter AC limits, and target.

Independent confirmation: a second Solinteg installation reproduced the mode-before-register requirement and the deceptive readback behavior. Source: <https://github.com/hspolander/solinteg-controller-oss>

Controller implication: on every full qEMS activation/relatch, do not reuse or trust prior BattCtrl register state. Enter EMS BattCtrl first, confirm it, then write the complete desired priority/limit/target set. When leaving qEMS, YOUEMS reasserts unrestricted EMS AC limits while qEMS is inactive so a restrictive profile is not intentionally carried into another inverter mode.

Same-mode handoff

When the controller is already active, the inverter is still in EMS BattCtrl, and the requested qEMS sub-mode is unchanged, perform no inverter initialization writes.

- Do not rewrite Working Mode.
- Do not rewrite priority.
- Do not rewrite AC limits.
- Do not reset the live BattCtrl target.
- For Battery Charge or Battery Discharge, update only the requested-power setting and let the running feedback loop ramp from the current live target.

This avoids abrupt target jumps and prevents schedules, cards, or repeated automation calls from disturbing a stable controller.

External working-mode changes

The inverter's actual working mode is authoritative. A qEMS sub-mode is valid only while the inverter reports EMS BattCtrl.

- If another system changes the inverter to EMS General or another unsupported mode, stop the qEMS feedback loop.
- Clear the remembered qEMS sub-mode.
- Do not send any Modbus writes.
- Do not force EMS BattCtrl back on.

This prevents a control tug-of-war between qEMS, the inverter UI, another scheduler, or another Home Assistant automation.

Runtime target updates

A feedback-driven qEMS profile can update 50207 from live PV, battery, and grid readings. The validated standalone controller uses freshness checks, a post-write settling interval, a deadband, a maximum step size, and an ownership recheck so stale measurements or a concurrent working-mode change cannot produce an inappropriate target write.

Before calculating a dynamic target, require the feedback entities used by that profile to be fresh. In the current Home Assistant standalone implementation, freshness is evaluated from each entity's last_reported timestamp and a reading older than 30 seconds is treated as stale. This 30-second value is a controller safety policy, not an inverter requirement.

If required feedback is stale or unavailable, skip the dynamic target update rather than calculating from old data. Do not interpret a stale-feedback pause as a BMS fault.

After an actual 50207 target write, allow a 2-second settling period before making another feedback correction. This is separate from activation synchronization: activation waits for confirmed EMS BattCtrl mode, whereas runtime settling gives the inverter and Home Assistant measurements time to reflect a target change.

Write only when the new command differs meaningfully from the live target. In the current tested qEMS implementation, the minimum target change is 0.01 kW.

Ramp large target changes rather than jumping immediately. The current tested maximum target step is 2 kW per write, with a nominal 1-second feedback loop when no post-write settling pause is active.

Immediately before every dynamic 50207 write, re-check controller ownership: qEMS must still be active, the expected qEMS sub-mode must still be current, and the inverter must still report EMS BattCtrl. If ownership or working mode changed after the target was calculated, discard the write. This closes the race between calculation and a concurrent mode change.

Keep the controller active when the BMS clamps power; target error is expected and is not itself a controller fault. Do not require battery-model-specific BMS status or permission entities.

Profiles and continuous feedback

Not every qEMS profile requires continuous power-balance feedback. Fixed-command profiles can establish their intended battery behavior primarily from the BattCtrl target and AC limits, while feedback-driven profiles must continually adapt 50207 to live site conditions.

Feedback-driven profiles: Self-Consumption, Feed-In, PV Charge+House, and Battery Discharge. Their target depends on live demand after PV (D), so stale-feedback protection applies to their dynamic corrections.

Primarily fixed-command profiles: PV Charge requests -M, Battery Charge requests -R, and Battery Freeze requests 0 kW with both AC limits at zero. Once their target and limits are established, they do not need continuous D-based correction to express the policy. Normal ownership/mode supervision and BMS/inverter safety behavior still apply.

Stopping and restoration

For an intentional qEMS shutdown while qEMS still owns EMS BattCtrl, use an orderly neutralize-and-cleanup sequence so a restrictive profile is not left behind.

First neutralize the battery command by writing target 50207 = 0 kW while EMS BattCtrl ownership is still valid.

Then restore the inactive/open BattCtrl configuration used by the standalone controller: PV priority, maximum inverter AC output = +200 kW, and maximum inverter AC input = -200 kW. These are practical unrestricted integration settings, not physical inverter ratings; the inverter and BMS retain their real limits.

Only after target neutralization and cleanup should the controller mark the qEMS sub-mode inactive. This ordering prevents the controller's own ownership guard from blocking its cleanup writes.

A manual controller can additionally save the previous working mode, priority, limits, and target before starting. On an explicit manual stop it may restore those saved values. If restoring EMS BattCtrl, use the same mode-first, confirm, then-register sequence.

If an external working-mode change already stopped qEMS, do not attempt the neutralize/cleanup sequence and do not restore or write anything automatically; ownership has already been lost.

A scheduler may instead stop qEMS and return to unrestricted EMS General behavior without restoring a manual pre-state.

qEMS control profiles

The following names are external qEMS policies built on EMS BattCtrl. They are not native inverter working modes. All use PV priority.

Let D = demand after PV = battery power - grid power $\approx$ house load - PV power. Let R = requested positive battery power and M = configured maximum target.

qEMS profile	50207 target	AC input limit	AC output limit
Feed-In	$\max(D, 0)$	0 (block charging from grid)	Open
Self-Consumption	D	Open	Open
PV Charge	$-M$	0	Open
PV Charge+House	$\min(D, 0)$	0	Open
Battery Charge	$-R$	Open	Open
Battery Discharge	$\max(R, D, 0)$	0	Open
Battery Freeze	0	0	0

"Open" means a sufficiently high limit so the EMS limit is not expected to constrain normal operation. The tested Home Assistant implementation uses 200 kW as a practical unrestricted setting exposed by the integration; the inverter and BMS still enforce their own real limits.

qEMS Feed-In

Purpose: export available PV instead of charging the battery, while using battery discharge only to avoid grid import.

- PV supplies the house first and remaining PV is exported.
- If PV is below house load, the battery covers the deficit when permitted.
- The battery is never intentionally charged.
- AC input is blocked.

qEMS Self-Consumption

Purpose: minimize site grid exchange.

- PV supplies the house first.
- PV surplus charges the battery.
- Battery discharge covers a remaining house deficit.
- Both inverter AC input and output are allowed.
- The desired result is near-zero grid flow, subject to BMS and inverter limits.

qEMS PV Charge

Purpose: give battery charging priority over immediate house use.

- The controller continuously requests the configured maximum charging target.
- Grid charging of the battery is blocked.
- Available PV is directed toward the battery first.
- The house is supplied from the grid to the extent needed while PV is reserved for charging.
- Actual charge power is limited by PV availability, inverter conversion limits, SOC, and BMS limits.
- **Important: blocking inverter AC input with 50209 = 0 does NOT make PV Charge a site-level “no grid import” mode. PV can be reserved for battery charging while the house simultaneously imports from the grid. This is expected behavior, not a fault. Use qEMS PV Charge+House when the intended policy is house-first solar use with only surplus PV charging the battery.**

qEMS PV Charge+House

Purpose: conventional house-first solar charging.

- PV supplies the house first.
- Only measured PV surplus is requested as battery charging power.
- The battery is not intentionally discharged.
- Grid charging is blocked.
- When PV is insufficient, the remaining house load comes from the grid.

qEMS Battery Charge

Purpose: charge the battery at a selected fixed target.

- 50207 is set to -R.
- PV contributes first under PV priority.
- Grid input supplies the remaining battery charge request and the house load.
- The BMS may reduce or reject the request without stopping the controller.

qEMS Battery Discharge

Purpose: guarantee at least the selected discharge request while still avoiding grid import.

- The requested value is a minimum discharge target, not a maximum.
- If the house deficit after PV is greater than R, the target rises to cover the deficit.
- If R is greater than the house deficit, excess battery power is exported.
- Battery charging and inverter AC input are blocked.

qEMS Battery Freeze

Purpose: keep the battery untouched.

- 50207 target is 0 kW.
- Maximum inverter AC output is 0.
- Maximum inverter AC input is 0.
- Confirmed testing held the battery close to zero power.
- Small transients can occur while the inverter settles or measurements update.

PV, house, and grid handling while both AC limits are zero can depend on firmware and installation topology, so verify live flows on each installation.

Practical examples with corrected 50207 semantics

Self-consumption with a house deficit

PV = 3 kW
 House load = 5 kW
 $D = \text{house} - \text{PV} = 2 \text{ kW}$
 Target = +2 kW

- Battery discharges approximately 2 kW.
- Grid flow approaches 0 kW if the battery can meet the request.

Feed-In with PV surplus

PV = 5 kW
 House load = 2 kW
 $D = -3 \text{ kW}$
 Target = $\max(D, 0) = 0 \text{ kW}$
 AC input limit = 0

- Battery charging is prevented.
- PV supplies the house and approximately 3 kW is exported.

Battery Charge at 4 kW

Requested battery charge $R = 4 \text{ kW}$
 PV = 3 kW
 House load = 0.7 kW
 Target = -4 kW

- PV contributes approximately 3 kW to battery charging.
- Grid contributes approximately 1 kW to battery charging.
- The house imports approximately 0.7 kW.
- Utility-meter import is therefore approximately 1.7 kW.

Battery Discharge at a minimum 4 kW

Requested discharge $R = 4 \text{ kW}$
 House deficit after PV $D = 1 \text{ kW}$
 Target = $\max(R, D, 0) = 4 \text{ kW}$

- Approximately 1 kW supplies the remaining house load.
- Approximately 3 kW is exported.

If D later rises to 6 kW, the target rises to approximately 6 kW to avoid grid import.

Battery Freeze

Target = 0 kW
 Max inverter output = 0 kW
 Max inverter input = 0 kW

- The controller requests neither charging nor discharging.
- Observed battery power remains close to zero after settling.

PV Charge compared with PV Charge+House

PV = 5 kW
 House load = 2 kW
 Maximum target $M = 10 \text{ kW}$

PV Charge: target = -10 kW, inverter AC input blocked for battery charging. The battery receives available PV first and the house can still import approximately 2 kW from the grid. This is not a site-level no-import mode.

PV Charge+House: target = -3 kW, inverter AC input blocked for battery charging. PV supplies the house first and only the 3 kW surplus charges the battery; the house imports only if PV is insufficient for house demand.

Status indications

EMS CmdLim

The requested battery power cannot be achieved because an EMS AC limit is constraining operation.

- Maximum inverter output reached.
- Maximum inverter input reached.
- Combined PV and battery power exceeds permitted inverter exchange.

PV PLim

PV has been curtailed.

- Maximum inverter output is below available PV.
- Battery priority is selected and battery power consumes available AC capacity.
- Battery cannot absorb surplus PV and export is restricted.

BMS-limited operation

A difference between requested and actual battery power is normal when the BMS applies SOC, temperature, voltage, or current limits. The qEMS controller does not treat this as a fault and does not require battery-model-specific readings.

Home Assistant entity mapping used by the standalone qEMS controller

Function	Entity ID
Working mode	select.solinteg_inverter_working_mode
Battery target (50207)	number.solinteg_inverter_battery_charge_discharge_power_target
Max AC output (50208)	number.solinteg_inverter_ems_battctrl_max_ac_power_limit
Max AC input (50209)	number.solinteg_inverter_ems_battctrl_min_ac_power_limit
Priority (50210)	select.solinteg_inverter_ems_battctrl_priority_of_power_output
PV power	sensor.solinteg_inverter_pv_power_total
Battery power	sensor.solinteg_inverter_battery_power
Native grid meter	sensor.solinteg_inverter_meter_active_power

Scope and test basis

These observations and control profiles are based on:

- Solinteg MHT firmware 9.x.
- EMS BattCtrl working mode.
- Native inverter metering.
- Testing with a Pylontech high-voltage battery.
- No EPS/off-grid load connected.
- Confirmed qEMS Self-Consumption, Battery Charge, Battery Discharge, and Battery Freeze behavior.

The qEMS implementation intentionally does not rely on battery-model-specific status entities. The inverter and BMS remain responsible for safety enforcement.

Validation reminder

Firmware revisions and installation topology can change edge behavior. Verify sign conventions, limits, and live power flow in dry-run or low-power testing before enabling unrestricted automation.

Acknowledgements and license

Independent validation credit. Several 2026-08-07 documentation clarifications were independently tested/reviewed by <https://github.com/hspolander/solinteg-controller-oss>, including the mode-before-register write-order rule, the fact that pre-mode values may read back without being enforced, the inverter-side meaning of registers 50208/50209, and the PV Charge house-import behavior. The 2026-08-08 revision also consolidates runtime hardening developed in the standalone qEMS controller: feedback freshness checks, post-target settling, pre-write ownership revalidation, feedback-driven versus fixed-command profile behavior, and orderly shutdown cleanup.

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